

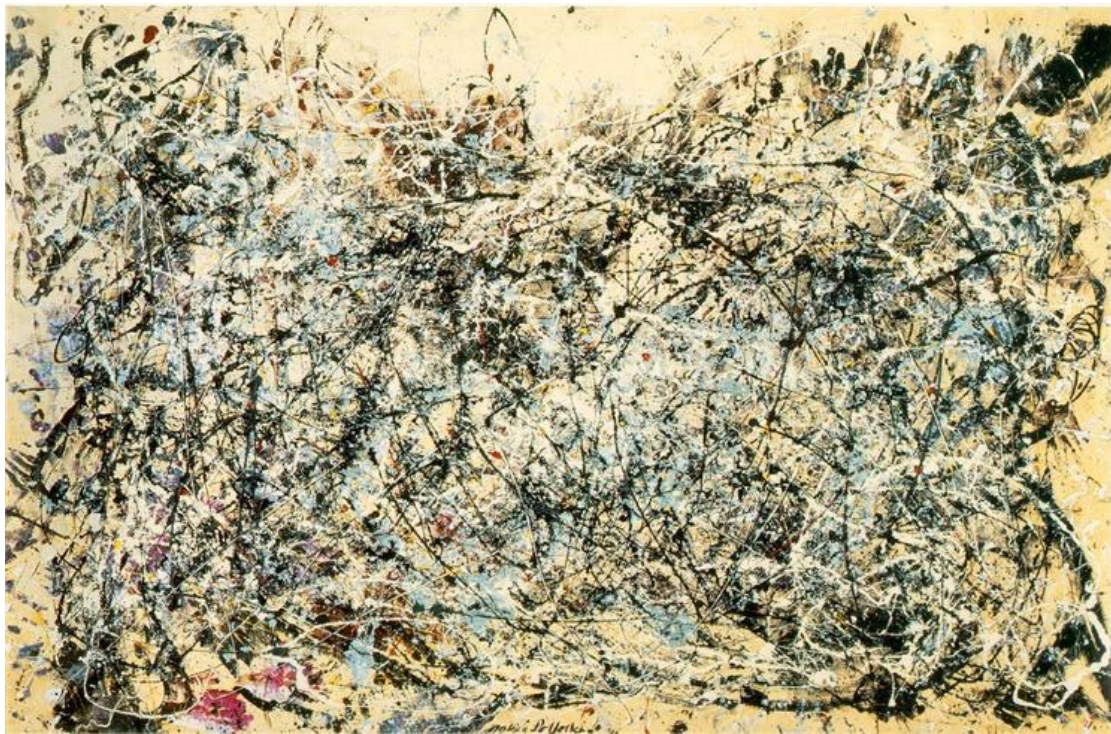
AI in Finance

Arpit Gupta (NYU Stern)
Spring 2026
Session 6

Portfolio Management and Trading



Signal and Noise



Number 1A, Jackson Pollock 1948

What makes a Pollock Pollock: a machine vision approach

Lior Shamir

Published Online: February 10, 2015 · pp 1-10 · <https://doi.org/10.1504/IJART.2015.067389>

PDF

Tools

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Abstract

Jackson Pollock introduced a revolutionary artistic style of dripping paint on a horizontal canvas. Here we study Pollock's unique artistic style by using computational methods for characterising the low-level numerical differences between original Pollock drip paintings and drip paintings of other painters who attempted to mimic his signature drip painting style. Four thousands and twenty four numerical image content descriptors were extracted from each painting, and compared using weighted nearest neighbour classification such that the Fisher discriminant scores of the content descriptors were used as weights. In 93% of the cases, the computer analysis was able to differentiate between an original and a non-original Pollock drip painting. The most discriminative image content descriptors that were unique to the work of Pollock were the fractal features, but other numerical image content descriptors such as Zernike polynomials, Haralick textures, and Chebyshev statistics show substantial differences between original and non-original Pollock drip paintings. These experiments show that the uniqueness of Pollock's drip painting style is not reflected merely by fractals, but also by other numerical image content descriptors that reflect the visual content. The code and software used for the experiment is publicly available, and can be used to study the work of other artists.

Scaling for Financial Trading

The Trading Workflow

- Traditional portfolio selection:
 - Estimate expected returns
 - More quant version: as function of factors or characteristics
 - Portfolio optimization
 - Curse of dimensionality
 - Translate model output into trades
 - Deal with transaction costs, market impact, other constraints

Challenges with AI Adoption

- Accuracy is crucial
 - Hard to rely on tools which might hallucinate or provide false information
- Hard to obtain right training data
 - Most open-source material doesn't fit financial application needs
 - Survival and look-ahead bias
- Deal with multiple modalities (pdfs, images, spreadsheets, etc)
- Building the right workflow incorporating AI components
- Non-stationarity: world is a moving object
- Adversarial Dynamics dilute advantages
- Asymmetric costs of being right vs being wrong
 - Varies by context; i.e. debt vs venture capital

Scaling Laws in Finance

Compute, Complexity, and the Scaling Laws of Return Predictability

Allan Timmermann Luka Vulicevic
UC San Diego *UC San Diego**

February 21, 2026

Abstract

Investors’ computing power (“compute”) governs how much signal they can extract from high-dimensional data. We show that forecast performance follows scaling laws—stable power-law relationships between accuracy and training compute—that pin down both an irreducible bound on return predictability and the rate at which additional compute improves accuracy. In firm-level cross-sectional return prediction, scaling laws explain over 80% of performance variation across models. Treating compute as a primitive yields sharp economic implications: scaling laws quantify predictability limits and the value of new data, distinguish problems with strong versus weak returns to scale, and deliver a market-efficiency metric by mapping computational advantage into a certainty equivalent. Measured this way, computational superiority earns sizable rents: a 25% increase in compute would have raised an investor’s Sharpe ratio by about 10% over the last 30 years, indicating substantial returns to computational sophistication.

Table A1: Public statements emphasizing compute and/or data scale.

Firm (linked)	Public statement (verbatim excerpt)
Renaissance Technologies	“A research database that grows by more than 40 terabytes a day ”; “ 52,000 computer cores ”; “ Redundant computational facilities .”
Hudson River Trading	“Researchers have access to massive compute, custom tooling ...”; “ high compute-per-researcher ratio ”
Two Sigma	“Store 300+ petabytes of data”; “Conduct more than 100,000 market data simulations daily”; “processing power measured in zettaFLOPS .”
Jane Street	“We work with petabytes of data, a computing cluster with hundreds of thousands of cores , and a growing GPU cluster containing thousands of high-end GPUs .”
Citadel Securities	“turn cutting-edge ideas and petabyte-scale data into ...models with direct trading impact.”; “Leverage large scale compute and data (petabytes; large budgets) to run ambitious experiments and push the frontier ”
Jump Trading	“Real-time inference on petabytes of market and alternative data”; “Massive HPC grid with thousands of GPUs ... ”; “ 5M+ Simulations Run Daily”
XTX Markets	“serve hundreds of petabytes and more than 100,000 compute nodes ”; “ 650 Petabytes of usable storage”; “ 25,000 GPUs in our research cluster”
D. E. Shaw	“...designs and maintains the firm’s largest compute infrastructure ...high-performance computing , networking, and storage for research and trading.”

Double Descent in Theory and Practice

Figure 1: Modern Machine Learning

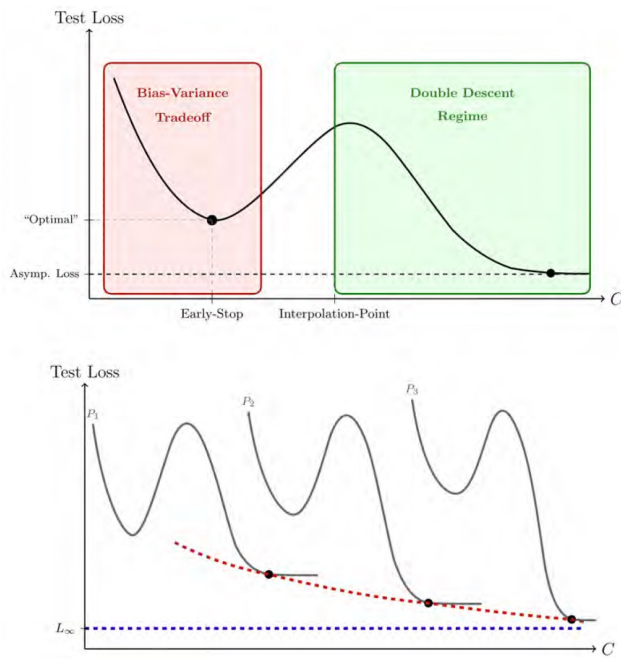
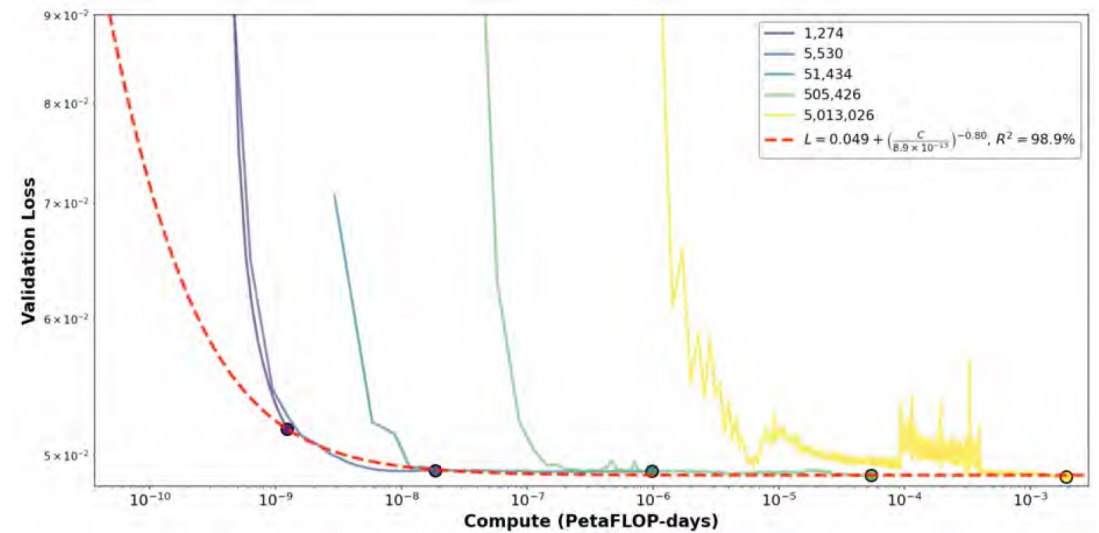


Figure A2: Double Descent



Note. This figure plots validation MSE against cumulative compute (PetaFLOP-days) for a subset of FNW36 firm-level return prediction networks. Colored lines trace validation loss over training for five model sizes (parameter counts in the legend); circled markers indicate each run's best validation loss, and the dashed red curve is the fitted scaling law. The non-monotone trajectories illustrate double descent around the interpolation regimes. Over-parameterized models must be trained sufficiently long enough (no early stopping) to pass the interpolation region and exploit implicit regularization that yields lower and more stable validation loss.

Financial Returns Follow Scaling Laws with More Data and Parameters

Figure 3: Scaling Laws of Firm-Level Return Prediction

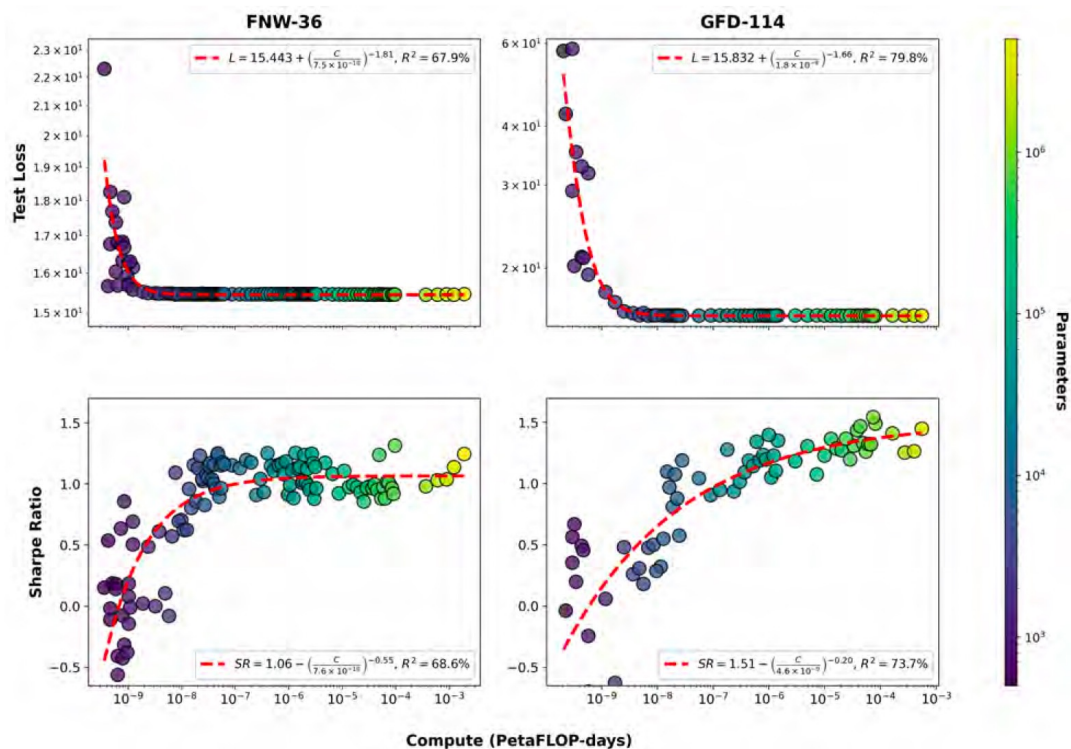
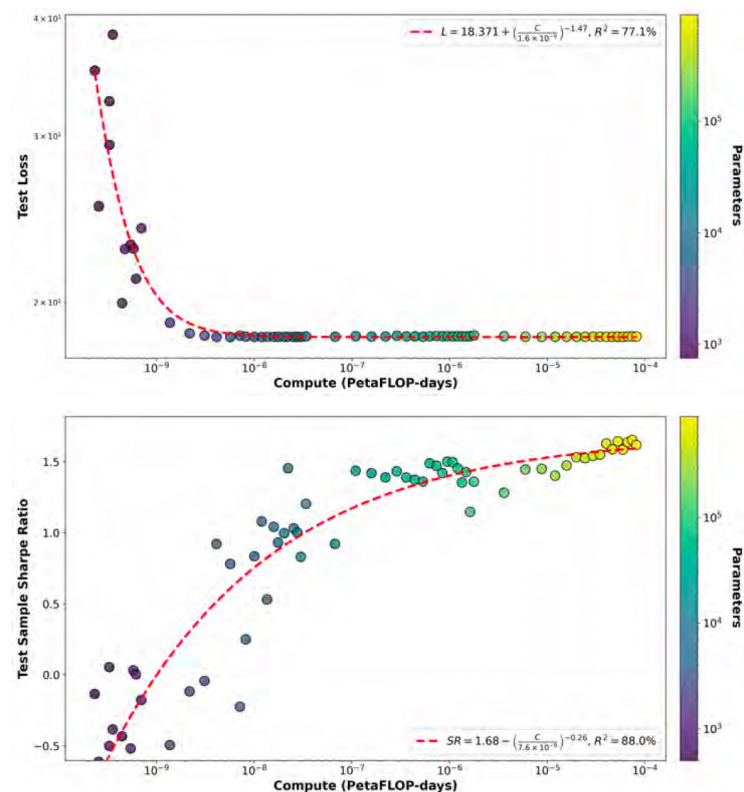
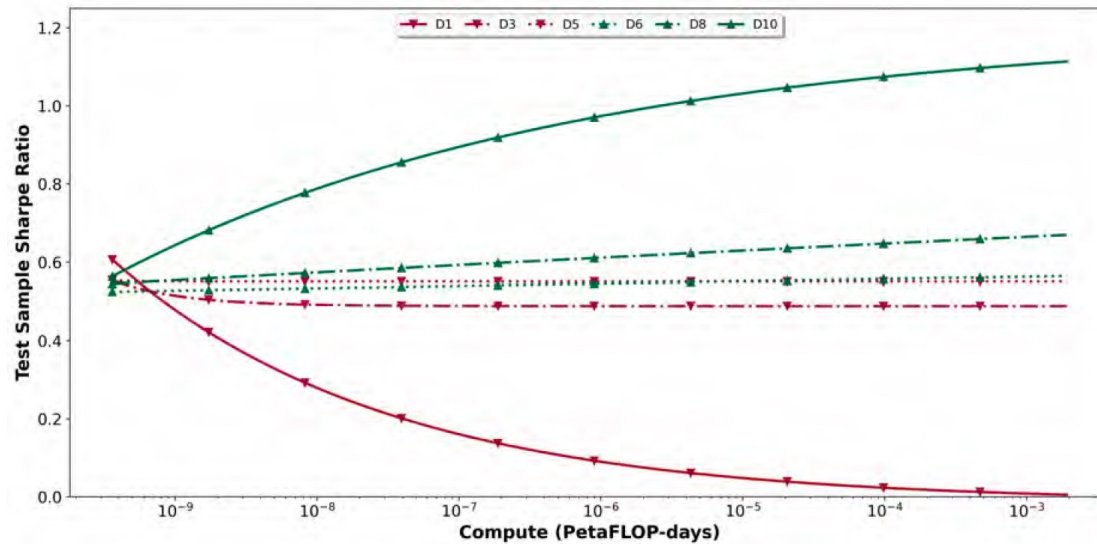


Figure 2: Firm-Level Return Scaling Laws (1994 - 2024)



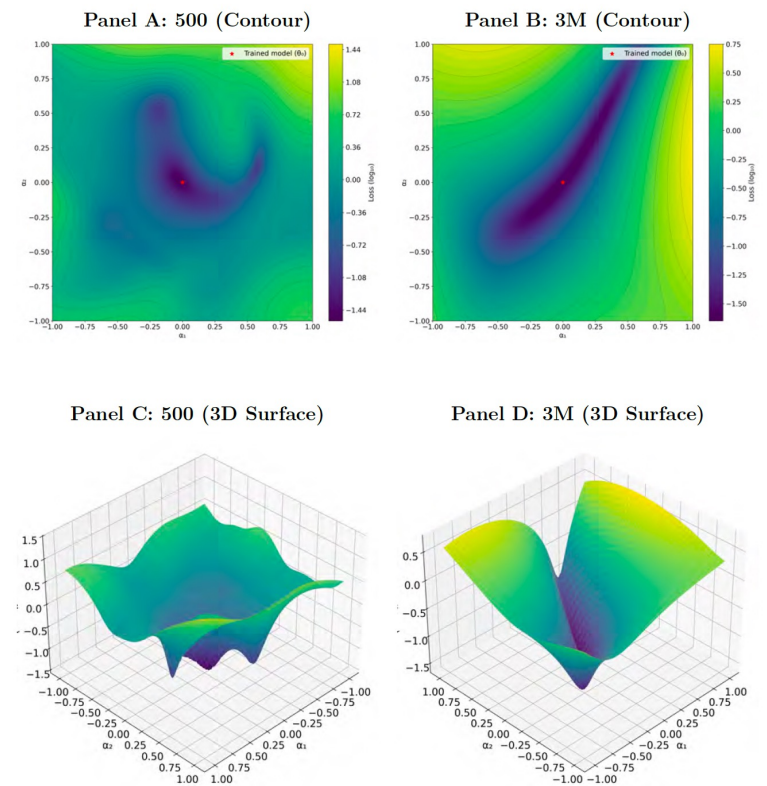
Extremes drive returns

Figure 6: Sharpe Ratio Scaling Laws by Decile



Note. This figure plots the annualized Sharpe ratios of the underlying decile portfolios from Figure 7.

Figure 9: Firm-Level Return Loss Landscapes Across Model Size



How AI Enables Increase in Scale

- What's new here?
 - “standard” models (linear or logistic regression) involve simple model, a few parameters
 - ML increases the number of parameters, but uses **regularization** to avoid overfitting
 - I.e., “lasso” puts weight of zero on a variable which does not contribute much to model fit
 - Empirical Bayes pulls model closer to “prior” or base rate
 - AI (ie, neural net) instead keeps weights on all variables, and finds unique combinations and interactions of variables in ways that allow improvements in model fit as data and model size increase

Limits to Scale

- Scaling laws hit diminishing returns as complexity grows
 - Initial returns high because it's easy to monetize better signals
 - Becomes an arms race among a few sophisticated players
- Unlike other domains, there are complications in finance:
 - **Trading costs** diminish returns beyond what back-testing suggests
 - Function of liquidity, order book depth, shorting costs
 - Over time, competitors may learn and mimic your strategy
 - World in which everyone has a complicated strategy hard to model!
 - Trying to hit a moving target: world is non-stationary, keeps changing

Living with Scaling Rules

- Many firms with compute-intensive strategy (ie RenTech) operate at low scale in AUM
 - Famously, RenTech keeps high value, hard to scale strategies in Medallion fund with internal capital, and outsider capital in other funds like Rief
 - Adversarial competitive responses limit gains from compute
- Alpha gets competed away; and scarce complements become data, infrastructure, and latency

Machine Predictions

Improvements in Forecasting Large Growth Stocks

Where Has All the Data Gone?

Maryam Farboodi, Adrien Matray, Laura Veldkamp, and Venky Venkateswaran

NBER Working Paper No. 26927

April 2020

JEL No. G14

ABSTRACT

As financial technology improves and data becomes more abundant, do market prices reflect this data growth? While recent studies documented rises in the information content of prices, we show that, across asset types, there is data divergence. Large, growth stock prices increasingly reflect information about future firm earnings. This is the rise reflected in the previous studies. But over the same time period, the information content of small and value firm prices was flat or declining. Our structural estimation allows us to disentangle these informational trends from changing asset characteristics. These facts pose a new puzzle: Amidst the explosion of data processing, why has this data informed only the prices of a subset of firms, instead of benefiting the market as a whole? Our structural model offers a potential answer: Large growth firms' data grew in value, as big firms got bigger and growth magnified the effect of these changes in size.

5 Conclusions

The ability of financial prices to transmit information about firm profitability is often regarded a measure of the efficiency or success of financial markets. As data processing technologies have improved, one might expect this to increase in the information content of all prices. Instead, what we find is that price informativeness has diverged across assets. While prices of large, growth stocks are getting more informative, the extent of information in prices of all other assets is stagnant or declining. This could reflect changes in information or changing asset characteristics. To disentangle trends in various factors influencing informativeness, we estimate a structural model and decompose the traditional measure of informativeness into changes in growth, volatility and information. This decomposition only strengthens the puzzle: Investors seem to be processing more and more data about large growth assets, but not about others.

To explore why data processing might diverge, we use the estimated structural model to impute a marginal value of data. We find that the value of large growth firm data has increased, primarily because these firms grew larger. Larger firms are more valuable to learn about, particularly if they are also expected to grow faster. This offers a partial explanation for the diverging trends in the information content of prices – investors chose to analyze the most valuable data, that pertaining to large, growth firms.

Figure 1: **Price Informativeness is Falling (Rising) for all Public Firms (S&P 500 Firms)**. The plots show the trends in price informativeness, estimated using (3), along with 95% confidence interval based on Newey-West standard errors with 5-lags. The left figure contains S&P 500 nonfinancial firms, while the right figure contains all publicly listed nonfinancial firms excluding S&P 500 firms.

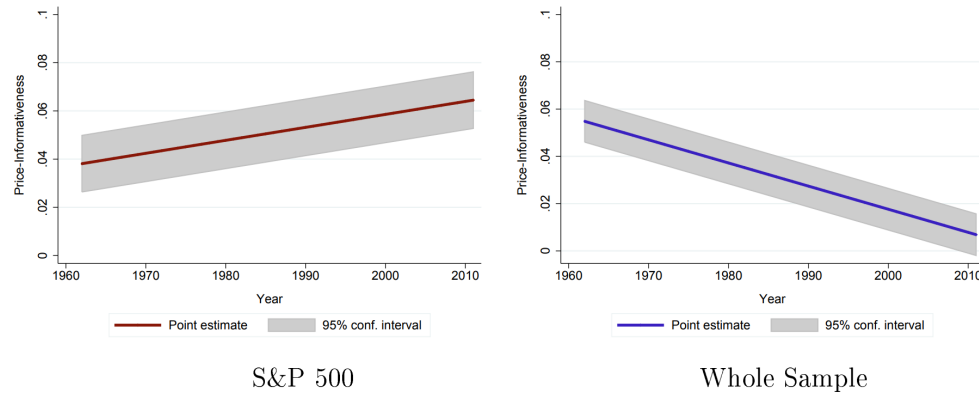
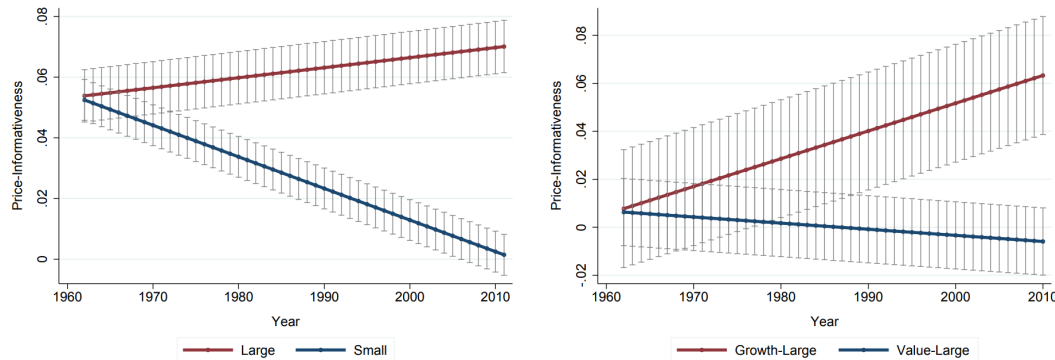


Figure 2: **Large and Small Firms' Price Informativeness Diverges.** The plots show the trends in price informativeness for horizon $s = 5$, estimated using (3), along with 95% confidence interval based on Newey-West standard errors with four year lags. Large refers to the 500 largest firms in our data – the rest are labeled Small. Growth firms are those in the bottom 30% of the distribution of book-to-market; value firms are in the top 30%.



What Are Factors?

- Factors are characteristics of stocks that predict returns
- You've heard of some already:
 - CAPM beta (market risk)
 - Book-to-market and E/P ratios (value investing)
 - Price momentum (trend following)
- The problem: Researchers have published 300+ factors
- The question: Which ones actually work out of sample?

The Factor Zoo in Practice

- Industry finds when tested rigorously, most factors fail
 - Multiple testing / p-hacking & microcap bias
 - Many factors are highly correlated
- ML solution is to let algorithms discover factor combinations
 - Key insight: It's not which factors, but how you combine them
 - ML improves out-of-sample forecasts vs. linear benchmarks

AI/ML Tools for Factor Selection

Empirical Asset Pricing via Machine Learning
Shihao Gu, Bryan Kelly, and Dacheng Xiu
NBER Working Paper No. 25398
December 2018, Revised September 2019
JEL No. C45, C55, C58, G11, G12

ABSTRACT

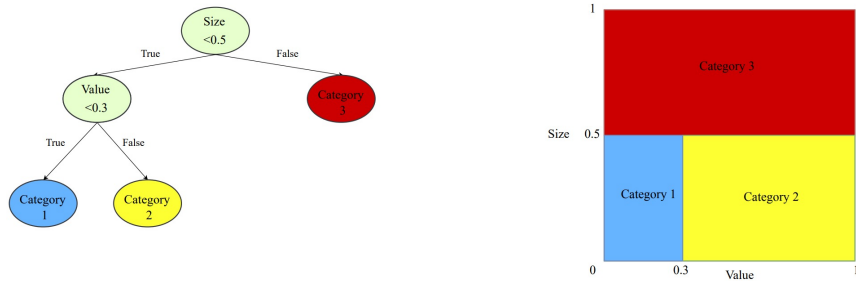
We perform a comparative analysis of machine learning methods for the canonical problem of empirical asset pricing: measuring asset risk premia. We demonstrate large economic gains to investors using machine learning forecasts, in some cases doubling the performance of leading regression-based strategies from the literature. We identify the best performing methods (trees and neural networks) and trace their predictive gains to allowance of nonlinear predictor interactions that are missed by other methods. All methods agree on the same set of dominant predictive signals which includes variations on momentum, liquidity, and volatility. Improved risk premium measurement through machine learning simplifies the investigation into economic mechanisms of asset pricing and highlights the value of machine learning in financial innovation.

4 Conclusion

Using the empirical context of return prediction as a proving ground, we perform a comparative analysis of methods in the machine learning repertoire. At the highest level, our findings demonstrate that machine learning methods can help improve our empirical understanding of asset prices. Neural networks and, to a lesser extent, regression trees, are the best performing methods. We track down the source of their predictive advantage to accommodation of nonlinear interactions that are missed by other methods. We also find that “shallow” learning outperforms “deep” learning, which differs from the typical conclusion in other fields such as computer vision or bioinformatics, and is likely due to the comparative dearth of data and low signal-to-noise ratio in asset pricing problems. Machine learning methods are most valuable for forecasting larger and more liquid stock returns and portfolios. Lastly, we find that *all* methods agree on a fairly small set of dominant predictive signals, the most powerful predictors being associated with price trends including return reversal and momentum. The next most powerful predictors are measures of stock liquidity, stock volatility, and valuation ratios.

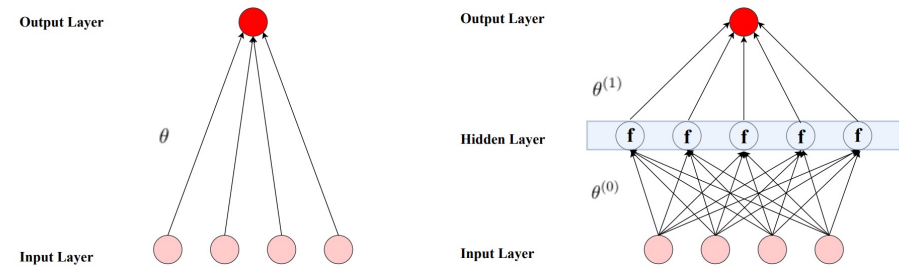
Tree and Neural Net Methods

Figure 1: Regression Tree Example



Note: This figure presents the diagrams of a regression tree (left) and its equivalent representation (right) in the space of two characteristics (size and value). The terminal nodes of the tree are colored in blue, yellow, and red, respectively. Based on their values of these two characteristics, the sample of individual stocks is divided into three categories.

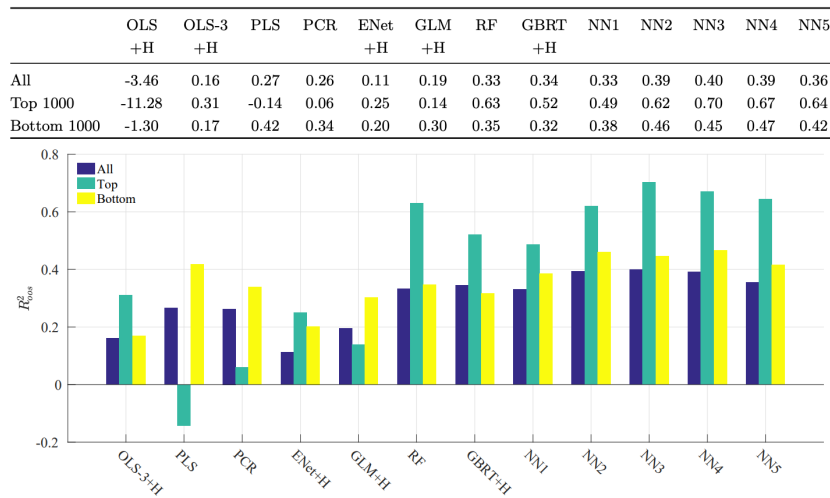
Figure 2: Neural Networks



Note: This figure provides diagrams of two simple neural networks with (right) or without (left) a hidden layer. Pink circles denote the input layer and dark red circles denote the output layer. Each arrow is associated with a weight parameter. In the network with a hidden layer, a nonlinear activation function f transforms the inputs before passing them on to the output.

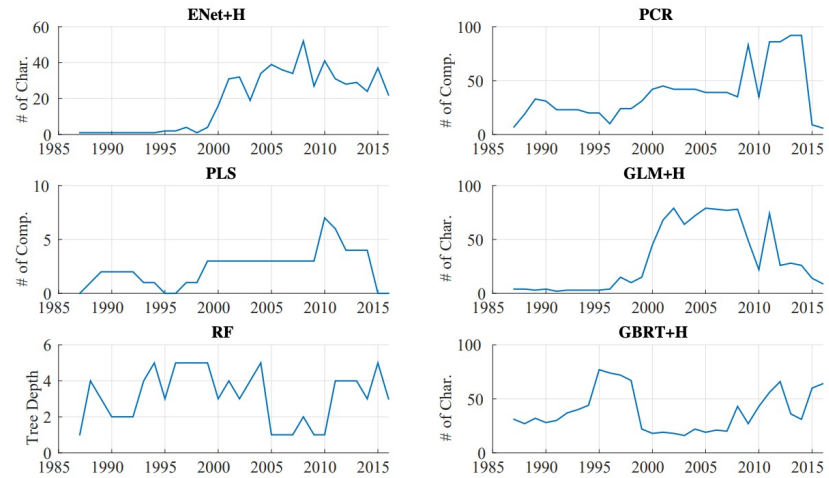
Taming the Factor Zoo

Table 1: Monthly Out-of-sample Stock-level Prediction Performance (Percentage R^2_{OOS})



Note: In this table, we report monthly R^2_{OOS} for the entire panel of stocks using OLS with all variables (OLS), OLS using only size, book-to-market, and momentum (OLS-3), PLS, PCR, elastic net (ENet), generalized linear model (GLM), random forest (RF), gradient boosted regression trees (GBRT), and neural networks with one to five layers (NN1–NN5). “+H” indicates the use of Huber loss instead of the l_2 loss. We also report these R^2_{OOS} within subsamples that include only the top 1,000 stocks or bottom 1,000 stocks by market value. The lower panel provides a visual comparison of the R^2_{OOS} statistics in the table (omitting OLS due to its large negative values).

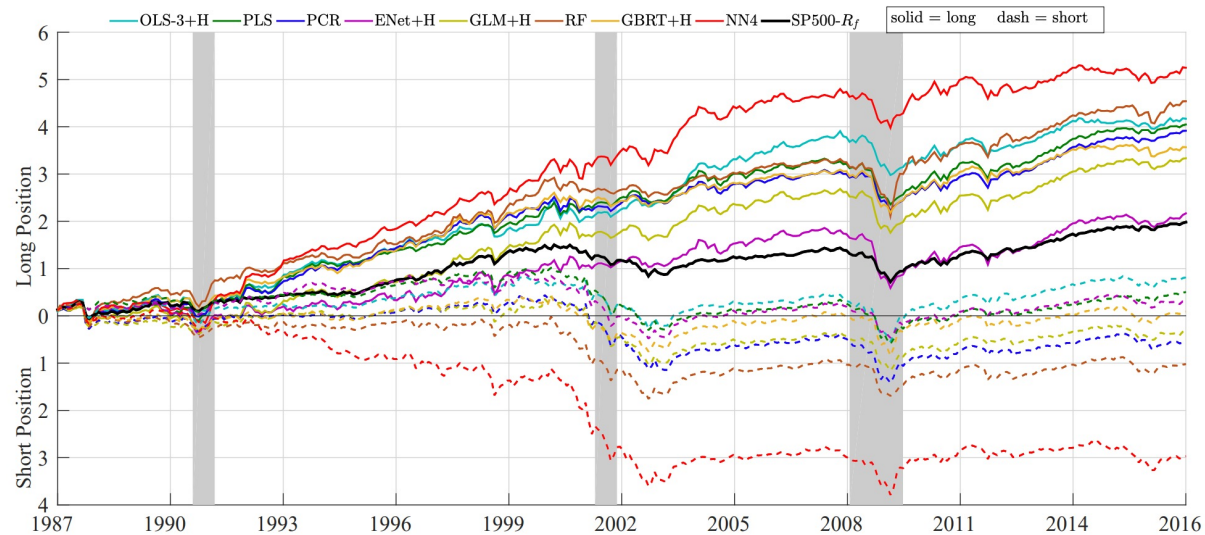
Figure 3: Time-varying Model Complexity



Note: This figure demonstrates the model complexity for elastic net (ENet), PCR, PLS, generalized linear model with group lasso (GLM), random forest (RF) and gradient boosted regression trees (GBRT) in each training sample of our 30-year recursive out-of-sample analysis. For ENet and GLM we report the number of features selected to have non-zero coefficients; for PCR and PLS we report the number of selected components; for RF we report the average tree depth; and for GBRT we report the number of distinct characteristics entering into the trees.

Long-Term Performance of ML Portfolios

Figure 9: Cumulative Return of Machine Learning Portfolios



Note: Cumulative log returns of portfolios sorted on out-of-sample machine learning return forecasts. The solid and dash lines represent long (top decile) and short (bottom decile) positions, respectively. The shaded periods show NBER recession dates. All portfolios are value weighted.

AI-Driven Image Analysis

(Re-)Imag(in)ing Price Trends

JINGWEN JIANG, BRYAN KELLY, and DACHENG XIU*

ABSTRACT

We reconsider trend-based predictability by employing flexible learning methods to identify price patterns that are highly predictive of returns, as opposed to testing predefined patterns like momentum or reversal. Our predictor data are stock-level price charts, allowing us to extract the most predictive price patterns using machine learning image analysis techniques. These patterns differ significantly from commonly analyzed trend signals, yield more accurate return predictions, enable more profitable investment strategies, and demonstrate robustness across specifications. Remarkably, they exhibit context independence, as short-term patterns perform well on longer time scales, and patterns learned from U.S. stocks prove effective in international markets.

of low-frequency data in financial markets, it would be beneficial to know if patterns that unfold at high frequencies (which we can potentially measure well) are similar to patterns that unfold at low frequencies.³ To illustrate this

³ This is motivated by the Mandelbrot (2013) “fractal” hypothesis in financial markets, which predicts that asset prices exhibit statistically self-similar patterns when studied at different time scales, and which also predicts the emergence of long-range dependence in prices (Cont (2005)).

Apparent Success of AI-Technical Analysis

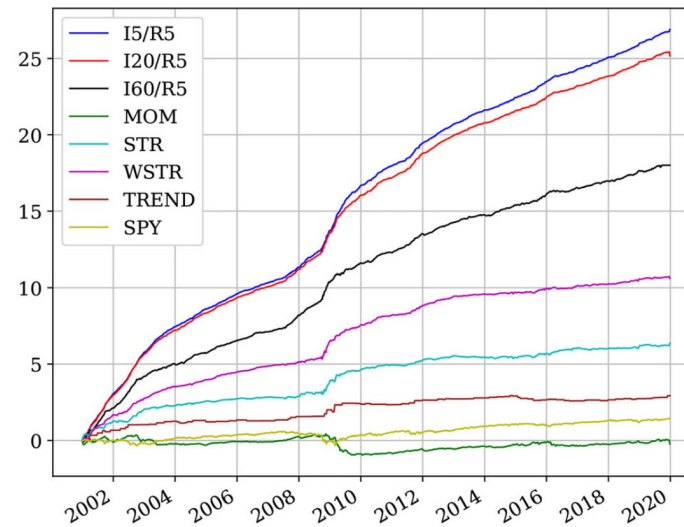


Figure 5. Cumulative volatility-adjusted returns of equal-weight portfolios. This figure displays cumulative volatility-adjusted log returns of equal-weight long-short portfolios of I5/R5, I20/R5, I60/R5, MOM, STR, WSTR, and TREND. All strategies are rescaled to have the same volatility as that of SPY over the test sample. (Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com))

Factor Zoo Strikes Back?

AI-Powered (Finance) Scholarship*

Robert Novy-Marx[†] Mihail Velikov[‡]

December 16, 2024

Abstract

This paper describes a process for automatically generating academic finance papers using large language models (LLMs). It demonstrates the process’ efficacy by producing hundreds of complete papers on stock return predictability, a topic particularly well-suited for our illustration. We first mine over 30,000 potential stock return predictor signals from accounting data, and apply the [Novy-Marx and Velikov \(2024\)](#) “Assaying Anomalies” protocol to generate standardized “template reports” for 96 signals that pass the protocol’s rigorous criteria. Each report details a signal’s performance predicting

Advantages of Complexity?

The Virtue of Complexity in Return Prediction
Bryan T. Kelly, Semyon Malamud, and Kangying Zhou
NBER Working Paper No. 30217
July 2022, Revised October 2022
JEL No. C1,C45,G1

ABSTRACT

Much of the extant literature predicts market returns with “simple” models that use only a few parameters. Contrary to conventional wisdom, we theoretically prove that simple models severely understate return predictability compared to “complex” models in which the number of parameters exceeds the number of observations. We empirically document the virtue of complexity in US equity market return prediction. Our findings establish the rationale for modeling expected returns through machine learning.

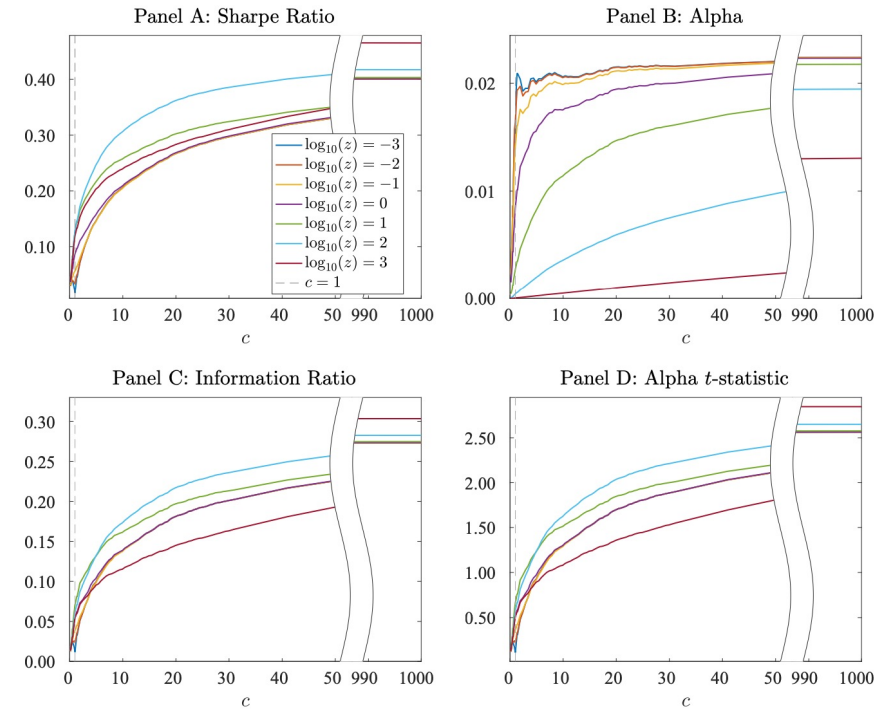


Figure 8: Out-of-sample Market Timing Performance ($T = 12$)

Note. Out-of-sample prediction accuracy and portfolio performance estimates for empirical analysis described in Section 6.3. Training window is $T = 12$ months and RFF count P (or cT) ranges from 2 to 12,000 with $\gamma = 2$. Alphas are versus a static position in the volatility-standardized market portfolio.

But Also Limitations

Seemingly Virtuous Complexity in Return Prediction *

Stefan Nagel[†]
*University of Chicago,
NBER, CEPR, and CESifo*

June 20, 2025

Return prediction with Random Fourier Features (RFF)—a very large number, P , of nonlinear transformations of a small number, K , of predictor variables—has become popular recently. Surprisingly, this approach appears to yield successful out-of-sample stock market index return predictions even when trained in rolling windows as small as $T = 12$ months with P in the thousands, and without shrinkage. However, this apparent virtue of complexity is an illusion. When $P \gg T$, the RFF-based predictive regression closely approximates a kernel ridgeless regression that takes the original K predictors as input. The resulting forecast is effectively a weighted average of past returns, with weights highest on periods whose predictor vectors are most similar to the current one. In short training windows, similarity simply means recency, so the forecast reduces to a weighted average of recent returns—essentially a momentum strategy. Moreover, similarity of current and past predictor vectors is decreasing in predictor volatility, which makes it a volatility-timed momentum strategy. Crucially, the volatility-timed momentum nature of the strategy arises mechanically from predictor similarity. The strong historical performance of the RFF-based strategy thus reflects the coincidental historical success of volatility-timed momentum, not predictive information extracted from training data.

Gaussian kernels. In this setting, return forecasts are constructed as weighted averages of the $T = 12$ lagged returns in the training window, where the weights depend on the distance between the current K -dimensional predictor vector and each lagged predictor vector. For example, if the one-month lagged predictor vector is more similar to the current one than the two-month lagged predictor vector, then the corresponding one-month lagged return receives a higher weight in the construction of the return forecast than the two-month lagged return. Because most predictor variables are persistent, recent predictor vector observations tend to be more similar to the current one, leading to higher positive weights on recent returns—producing a momentum-like forecast. Moreover, when the predictor variables are less volatile, distances between predictor vectors are smaller, resulting in higher weights. Thus, the strategy embeds a form of volatility timing, where the overall strength of the momentum signal varies inversely with predictor volatility in the training window. The combination of these effects yields a volatility-timed momentum strategy.

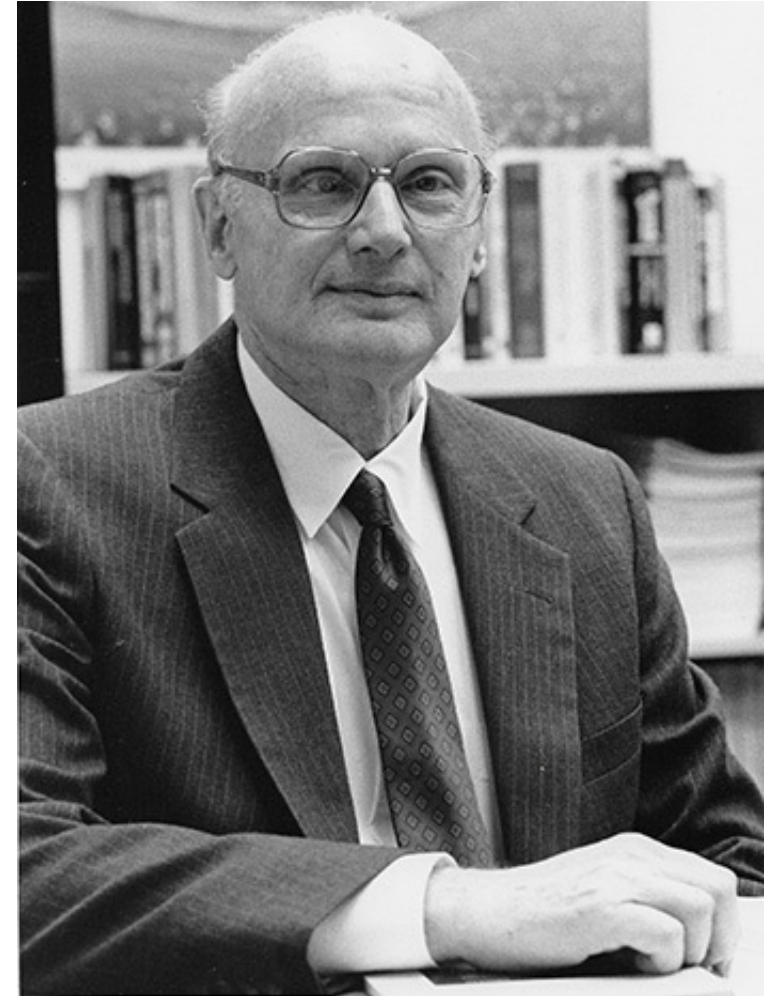
Economics of AI/ML for Signals

- ML/AI methods can improve return prediction
- But gains can be fragile, the signals decay with mass adoption
- Can be hard to figure out if new methods really “learn” new approaches
 - Is it finding real economic structure, or patterns that happen to work in-sample?
- Also easier to fabricate fake stories

Portfolio Selection

The Markowitz Problem

- N Assets, want to maximize returns for a given amount of risk
- You have the expected return μ and a covariance matrix Σ (ie, the variance of a stock and its covariance/correlation with all other stocks)
- Find the best combination of assets



The Curse of Dimensionality

- A covariance matrix for N assets has $N(N+1)/2$ unique parameters
- 10 assets -> 55 parameters
- 100 assets -> 5,050 parameters
- 500 assets (S&P 500) -> 125,250 parameters
- 3,000 assets (Russell 3000) -> 4.5 million parameters!
- To estimate this reliably, you need $T \gg N$. But with monthly data, 500 assets, and 20 years, you have 125,250 parameters.
- Michaud (1989) showed the mean-variance optimized portfolio is very brittle and not robust to estimation error

Very Hard to Beat Indexing

JOURNAL ARTICLE

Optimal Versus Naive Diversification: How Inefficient is the $1/N$ Portfolio Strategy?

[Get access >](#)

[Victor DeMiguel](#) , [Lorenzo Garlappi](#), [Raman Uppal](#)

The Review of Financial Studies, Volume 22, Issue 5, May 2009, Pages 1915–1953,
<https://doi.org/10.1093/rfs/hhm075>

Published: 03 December 2007

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Abstract

We evaluate the out-of-sample performance of the sample-based mean-variance model, and its extensions designed to reduce estimation error, relative to the naive $1/N$ portfolio. Of the 14 models we evaluate across seven empirical datasets, none is consistently better than the $1/N$ rule in terms of Sharpe ratio, certainty-equivalent return, or turnover, which indicates that, out of sample, the gain from optimal diversification is more than offset by estimation error. Based on parameters calibrated to the US equity market, our analytical results and simulations show that the estimation window needed for the sample-based mean-variance strategy and its extensions to outperform the $1/N$ benchmark is around 3000 months for a portfolio with 25 assets and about 6000 months for a portfolio with 50 assets. This suggests that there are still many “miles to go” before the gains promised by optimal portfolio choice can actually be realized out of sample.

What Goes Wrong?

- Correlations are not stable. They spike in the worst times, ie “All correlations are 1 in a crash.”
- Relationships might not be linear. Hard to catch threshold effects, non-linear dynamics, contagion dynamics etc.
- Past Performance is no guarantee for...

Asset Embeddings

Asset Embeddings*

Xavier Gabaix[†] Ralph S.J. Koijen[‡] Robert J. Richmond[§] Motohiro Yogo[¶]

August 1, 2025

Abstract

Traditional representations of firms use accounting and financial market data, but investors use richer information sets. Theoretically, portfolio holdings contain all relevant information for asset prices, recoverable under empirically realistic conditions. Building on recent advances in machine learning and artificial intelligence, we develop asset embeddings that leverage portfolio holdings to represent firms, similar to word embeddings leveraging document structure. We evaluate different methods of estimating asset embeddings on three new benchmarks. We also develop investor embeddings to represent investors and their strategies. We economically interpret asset (investor) embeddings by applying large language models to firm- (investor-)level text data.

JEL: C53, G12, G23

Keywords: Artificial intelligence, Machine learning, Asset pricing, Recommender systems, Transformer models, Benchmarks

"Masked" Portfolio Positions

Holdings Data - ARKK

As of 07/07/2023



ARKK

ARK Innovation ETF

	Company	Ticker	CUSIP	Shares	Market Value (\$)	Weight (%)
1	TESLA INC	TSLA	88160R101	3,496,872	\$967,024,982.88	12.43%
2	COINBASE GLOBAL INC -CLASS A	COIN	19260Q107	7,945,138	\$620,515,277.80	7.98%
3	ROKU INC	ROKU	77543R102	8,865,426	\$546,110,241.60	7.02%
4	ZOOM VIDEO COMMUNICATIONS-A	ZM	98980L101	8,258,591	\$594,248,251.99	6.87%
5	UIPATH INC - CLASS A	PATH	90364P105	28,152,366	\$463,106,420.70	5.95%
6	BLOCK INC	SQ	852234103	7,069,493	\$456,759,942.73	5.87%
7	EXACT SCIENCES CORP	EXAS	30063P105	4,031,264	\$368,739,718.08	4.74%
8	UNITY SOFTWARE INC	U	91332U101	8,350,868	\$338,627,697.40	4.35%
9	SHOPIFY INC - CLASS A	SHOP	82509L107	5,430,238	\$335,751,615.54	4.32%
10	DRAFTKINGS INC-CL A	DKNG UW	26142V105	12,035,607	\$303,658,364.61	3.90%

Figure 1. An Example of a Masked Portfolio. This figures shows the top ten holdings of the ARK Innovation ETF in July 2023. The fourth largest position, which is Zoom Video Communications, is masked by a red line.

Embeddings Generate Firm Similarity

Table 2. Text-Based Embeddings from OpenAI

Rank	Apple Inc	Citigroup Inc	Walmart Inc
1	Appian Corp	Citizens Financial Group Inc	Walgreens Boots Alliance Inc
2	Adobe Inc	Goldman Sachs Group Inc	Home Depot Inc
3	Interdigital Inc	American International Group Inc	Murphy USA Inc
4	Microsoft Corp	Comerica Inc	Amazon Com Inc
5	Gopro Inc	Cigna Corp New	Qurate Retail Inc
6	Netapp Inc	Capital One Financial Corp	Big Lots Inc
7	Intel Corp	Caci International Inc	Burlington Stores Inc
8	Alphabet Inc	Capital City Bank Group	Dollar Tree Inc
9	Autodesk Inc	C N O Financial Group Inc	Nordstrom Inc
10	Appfolio Inc	JP Morgan Chase & Co	Kohls Corp

This table reports the ten closest firms to Apple, Citigroup, and Walmart, according to the text-based embeddings from OpenAI.

8.2. Interpreting Asset Embeddings

We feed the earnings calls that took place during the first quarter of 2023 to the LLM. We prompt the LLM for each of the earnings call transcripts:

Analyze this earnings call transcript and provide a structured summary providing:

****Investor Appeal Factors****

- What makes this company attractive for portfolio inclusion
- Type of company (e.g. technology, financial services, etc.)
- Growth drivers and expansion opportunities
- Market trends the company is positioned to benefit from
- Risks to which the company is exposed and potential headwinds

Focus on extracting insights that would influence portfolio construction decisions rather than just reporting numbers. Don't mention company names or products. Refer to the company as "the company" and the products as "the products".

We remove company and product names to limit the model's reliance on information in its training set, although such knowledge may be a virtue if our focus is on interpretability and developing economic narratives. We then feed all summaries in the LLM using the prompt (in addition to formatting instructions):

Below are summaries of 10 earnings transcripts of companies that appear often together in the portfolios of investors. Provide the common investor appeal factors of the companies.

This analysis yields the following results.

Interpreting Embedding Similarities

Understanding the commonality in 10 firms most similar to Apple

- **Market Leadership and Dominant Positioning:** The companies are consistently described as leaders in their respective industries, possessing dominant market share, strong brand power, and valuable intellectual property or franchises.
- **Positioned for Major Secular Growth Trends:** A primary appeal is their strategic positioning to capitalize on durable, long-term market shifts. Artificial Intelligence (AI) is a near-universal theme, cited as a core growth driver and a new computing platform. Other key trends include the enterprise shift to cloud computing, digital transformation, and the growth of the digital/creator economy.
- **Financial Strength and Resilience:** They demonstrate robust financial health, often reporting record revenues, strong operating margins, and significant free cash flow generation, even in challenging macroeconomic environments.
- **Strong Customer Loyalty and Ecosystems:** Many of the companies have built powerful, sticky ecosystems that foster high customer loyalty and recurring revenue. This is evident through massive installed user bases, membership models with high renewal rates, and direct-to-consumer strategies.
- **Clear Growth and Innovation Strategies:** Each company has well-defined avenues for future growth, including aggressive investment in product innovation, expansion into new product categories or services, and a focus on international market penetration.
- **Commitment to Shareholder Returns:** A common factor is a stated commitment to returning capital to shareholders, frequently through significant share repurchase programs and, for some, the payment or reinstatement of dividends.

Embedding Similarities for Citigroup

Understanding the commonality in 10 firms most similar to Citigroup

- **Strong Shareholder Returns:** Nearly all companies emphasize a strong commitment to returning capital to shareholders through consistent dividends (often with a history of increases) and significant share repurchase programs.
- **Robust Financial Performance and Profitability:** A common theme is the delivery of strong financial results, including record or industry-leading earnings, robust cash flow, and expanding profit margins in core businesses.
- **Market Leadership:** The companies are frequently leaders in their respective industries or in highly profitable market segments, which provides a stable financial foundation and competitive advantage.
- **Strong Balance Sheet and Capital Position:** A key appeal is maintaining a strong or "fortress" balance sheet, characterized by high capital levels (e.g., CET1), low debt, and financial flexibility to navigate economic cycles and fund growth.
- **Clear Strategic Initiatives for Growth:** Each company presents a clear strategic plan for future growth, whether through transformation (e.g., pivoting to EVs, digital banking, or smoke-free products), operational turnarounds, strategic simplification, or investment in new high-growth areas like oncology and low-carbon solutions.
- **Positioned for Favorable Market Trends:** The companies are strategically positioned to capitalize on major secular and cyclical trends, such as rising interest rates, the energy transition, digital adoption, and shifts in consumer preferences.
- **Operational Excellence and Cost Discipline:** Many highlight successful cost-reduction programs, operational efficiency gains, and disciplined management as key drivers for improving profitability and offsetting headwinds like inflation.

Embeddings Explain Portfolio Holdings

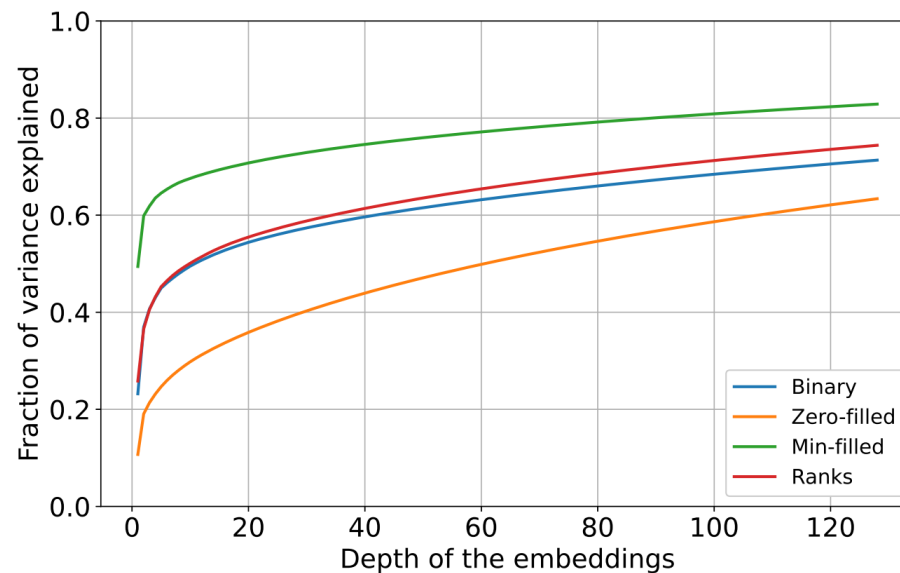


Figure 3. Variation in the Portfolio Holdings Explained by the Recommender Systems. The models are RS-Binary, RS-Ranks, RS-L-0, and RS-L-Min. See Table 1 for the model descriptions. The embedding dimension ranges from 4 to 128. The quarterly estimates for 2005.Q1 to 2022.Q4 are averaged over the entire sample period.

Embeddings Explain Asset Valuations

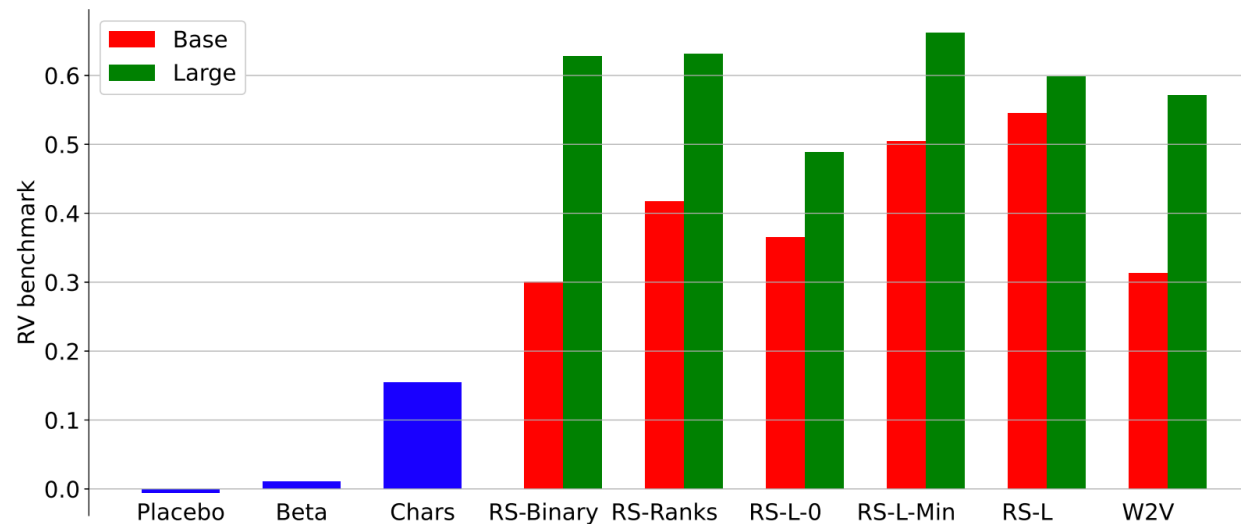


Figure 5. Relative Valuation Benchmark. From left to right, the models are placebo embeddings; the market beta as a characteristic; four characteristics (i.e., market beta, asset growth, profitability, and dividend-asset ratio); recommender systems RS-Binary, RS-Ranks, RS-L-0, RS-L-Min, and RS-L; and Word2Vec. See Table 1 for the model descriptions. The base model has four dimensions, and the large model has ten dimensions. The quarterly sample period is 2005.Q1 to 2022.Q4.

Deep Learning for Portfolio Optimization

Deep Learning for Portfolio Optimization

Zihao Zhang, Stefan Zohren, Stephen Roberts
Oxford-Man Institute of Quantitative Finance,
University of Oxford

Abstract

We adopt deep learning models to directly optimise the portfolio Sharpe ratio. The framework we present circumvents the requirements for forecasting expected returns and allows us to directly optimise portfolio weights by updating model parameters. Instead of selecting individual assets, we trade Exchange-Traded Funds (ETFs) of market indices to form a portfolio. Indices of different asset classes show robust correlations and trading them substantially reduces the spectrum of available assets to choose from. We compare our method with a wide range of algorithms with results showing that our model obtains the best performance over the testing period, from 2011 to the end of April 2020, including the financial instabilities of the first quarter of 2020. A sensitivity analysis is included to understand the relevance of input features and we further study the performance of our approach under different cost rates and different risk levels via volatility scaling.

4.3 Training Scheme

In this work, we use a single layer of LSTM connectivity, with 64 units, to model the portfolio weights and thence to optimise the Sharpe ratio. We purposely keep our network simple to indicate the effectiveness of this end-to-end training pipeline instead of carefully fine-tuning the “right” hyperparameters. Our input contains close prices and daily returns for each market index and we take the past 50 days of these observations to form a single input. We are aware that returns can be derived from prices, but keeping returns help with the evaluation of Equation 7 and we can also treat them as momentum features in [26]. As our focus is not on feature selection, we choose these commonly used features in our work. The Adam optimiser [17] is used for training our network, and the mini-batch size is 64. We take 10% of any training data as a separate validation-set to optimise hyperparameters and control overfitting problems. Any hyperparameter optimisation is done on the validation set, leaving the test data for the final performance evaluation and ensuring the validity of our results. In general, our training process stops after 100 epochs.

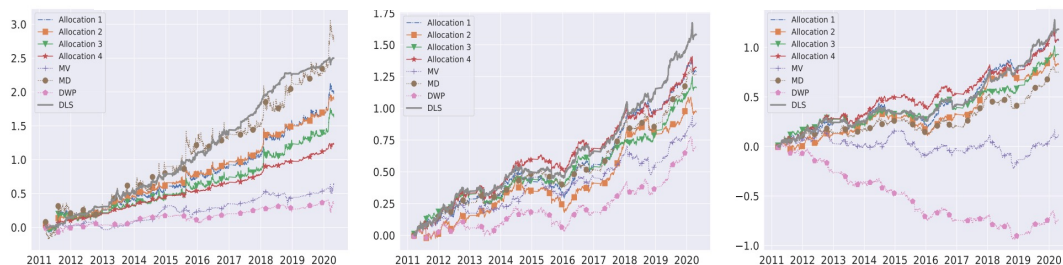


Figure 3: Cumulative returns (logarithmic scale) for **Left**: no volatility scaling and $C = 0.01\%$; **Middle**: volatility scaling ($\sigma_{tgt} = 0.10$) and $C = 0.01\%$; **Right**: volatility scaling ($\sigma_{tgt} = 0.10$) and $C = 0.1\%$.

Economies of Scale in Prediction

The fundamental law of active management

Mc² is the Law, here as well as elsewhere.

Richard C. Grinold

BR is the strategy's *breadth*. Breadth is defined as the number of independent forecasts of exceptional return we make per year.

IC is the manager's *information coefficient*. This measure of skill is the correlation of each forecast with the actual outcomes. We have assumed for convenience that IC is the same for all forecasts.

The law connects breadth and skill to the information ratio through the (approximately true) formula:

$$IR = IC \cdot \sqrt{BR} \quad (6.1)$$

Implications of Breadth

- If you have nine independent signals which predict returns, finding a tenth can be very valuable
 - Can “stack” all ten together, assuming they pay off separately, achieving even higher returns uncorrelated with aggregate or idiosyncratic risk
- Helps to explain:
 - Some scale economies among quant funds (Two Sigma, Citadel, etc.)
 - The “pod shop” model in hedge funds

Methods of Portfolio Construction

- **Asset Embeddings:** Instead of estimating all pairwise covariances, learn a low-dimensional vector representation of each asset
- **Hierarchical Risk Parity** (de Prado): Use clustering algorithms to group assets by similarity, then allocate risk across and within clusters
- **Deep Learning:** Use neural networks to map large, messy inputs — returns, fundamentals, text, macro data into forecasts of risk, return, or portfolio weights
- **Pod Shops/Quant Funds:** Layer multiple trading desks in parallel

Matching Human Behavior

AI can Mimic Human Mutual Fund Managers

Mimicking Finance

Lauren Cohen, Yiwen Lu, and Quoc H. Nguyen

NBER Working Paper No. 34849

February 2026

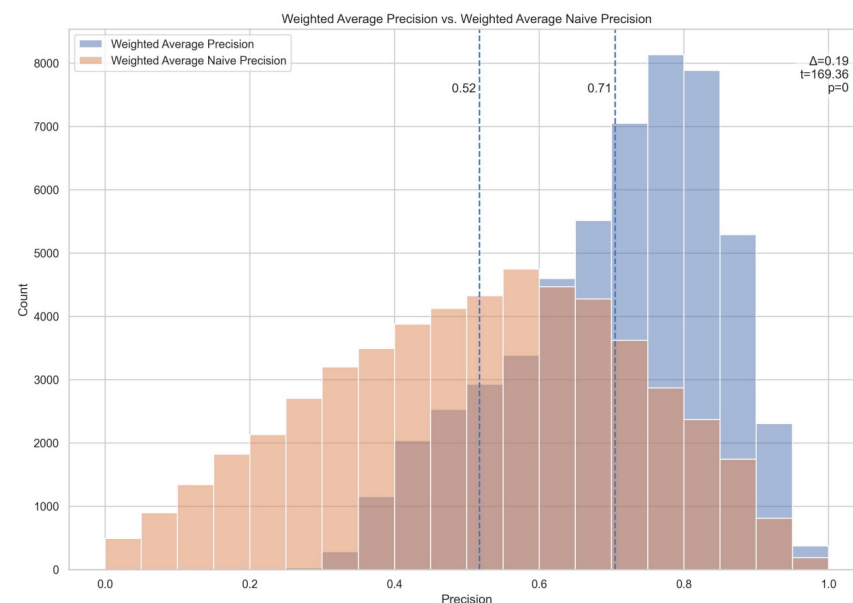
JEL No. C45, C53, C55, C82, G11, G23

ABSTRACT

We use frontier advancements in Artificial Intelligence and machine learning to extract and classify the part of key economic agents' behaviors that are predictable from past behaviors. Even the agents themselves might view these as novel (innovative) decisions; however, we show in strong contrast that a large percentage of these actions and behaviors can be predicted—and thus mimicked—in the absence of these individuals. In particular, we show that 71% of mutual fund managers' trade directions can be predicted in the absence of the agent making a single trade. For some managers, this increases to nearly all of their trades in a given quarter. Further, we find that manager behavior is more predictable and replicable for managers who have a longer history of trading and are in less competitive categories. The larger the ownership stake of the manager in the fund, the less predictable their behavior. Lastly, we show strong performance implications: less predictable managers strongly outperform their peers, while the most predictable managers significantly underperform. Even within each manager's portfolio, those stock positions that are more difficult to predict strongly outperform those that are easier to predict. Aggregating across the universe of fund managers each quarter, stocks whose position changes are least predictable additionally significantly outperform stocks whose position changes are most predictable across the universe. Our framework allows researchers to delineate and classify the portion of financial agents' action sets which are predictable from those which are novel responses to stimuli -- open to being evaluated for value creation or destruction.

Figure 3. Performance Precision

This figure compares the distribution of model-based *Weighted Average Precision* with the *Weighted Average Naive Precision*. The vertical dashed lines indicate the mean values.



Representing Financial Analyst Behavior

Mental Models and Financial Forecasts^{*}

Francesca Bastianello Paul H. Décaire Marius Guenzel[†]

November 22, 2025

Abstract

We uncover the mental models financial professionals use to explain their quantitative forecasts, and show how they shape beliefs and return predictability. Using the near-universe of 2.1 million equity analyst reports, we collect the valuation methods analysts adopt to compute their price targets, together with their reasoning, measured as attention to topics, and their associated valuation channels, time horizons, and sentiments. To validate the reliability of our output, we introduce a multi-step LLM prompting strategy and new diagnostic tools. Consistent with a model of top-down and bottom-up attention, we then uncover three sets of facts. First, analysts' mental models are sparse and rigid, and the choice of attention allocation and valuation methods are jointly determined by both analyst- and firm-characteristics. Second, analysts' reasoning translates into their quantitative forecasts. Both attention and valuation methods contribute to differences in valuations over time and across analysts, but variation in attention plays a bigger role. Third, we study the extent to which different topics contribute to over and underreaction to information, and show how biases in analysts' reasoning are reflected in asset prices. Analysts underreact to macroeconomic topics, and overreact to firm-related topics, and this contributes to return predictability.

Valuation Methods in Analyst Reports

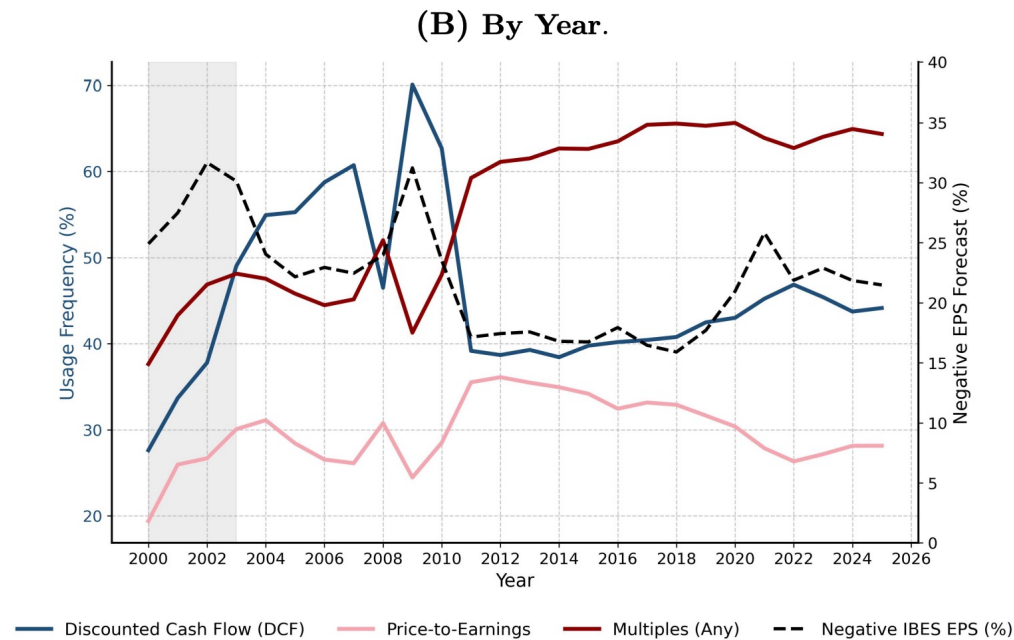


Figure III:
Valuation Methods

Topics Covered by Analysts



Figure VIII:
Topic Attention Allocation Over Time (Top 50 Topics)

Investor Agents in Silico

The Market’s Mirror: Revealing Investor Disagreement with LLMs*

Vineet Bhagwat[†]

J. Anthony Cookson[‡]

Chukwuma Dim[§]

Marina Niessner[¶]

January 2026

Abstract

Large language models (LLMs) can emulate human perspectives. Leveraging this idea, we study how investor disagreement emerges in response to firm news. We endow an LLM with 216 representative investor personas and elicit buy, hold, or sell responses to S&P500 firm news headlines from 2010-2025. Dispersion in responses yields article-level disagreement for 5.5 million headlines and sheds light on its sources. Disagreement is highest for socially-oriented news and lowest for fundamentals. Persona responses reflect non-pecuniary rationales and align with human-survey benchmarks. Disagreement predicts elevated same- and next-day abnormal trading volume, especially retail, and results persist beyond the LLM’s training cutoff.

Constructing Investor Personas

We use LLMs to fill these gaps:

- Our approach is to construct **AI personas**, using LLMs as a structural model of human perspective (studying disagreement in “silico”).
- Using LLMs, at scale, to generate an article-level disagreement measure tied to demographics.

Our approach

- First, we imbue a LLM (Llama 3.1 8B) with 216 “investor personas”
 - Investor personas: FINRA investor characteristics + politics
- Second, we ask all 216 personas to give their buy/hold/sell sentiment about each news headline & a justification.
 - **Sample frame:** S&P500 firms, 2010-2025; RavenPack DJNW headlines
 - Only news **about** firms.
 - Disagreement = weighted SD of buy/hold/sell

Key Findings Show Disagreement Along Politics and Income

We find headline-level disagreement across personas that makes contextual sense:

- Executive pay & trading, environmental issues, and government bailouts generate the most disagreement
- Persona “rationales” differ along efficiency grounds, fairness ground, and so on.

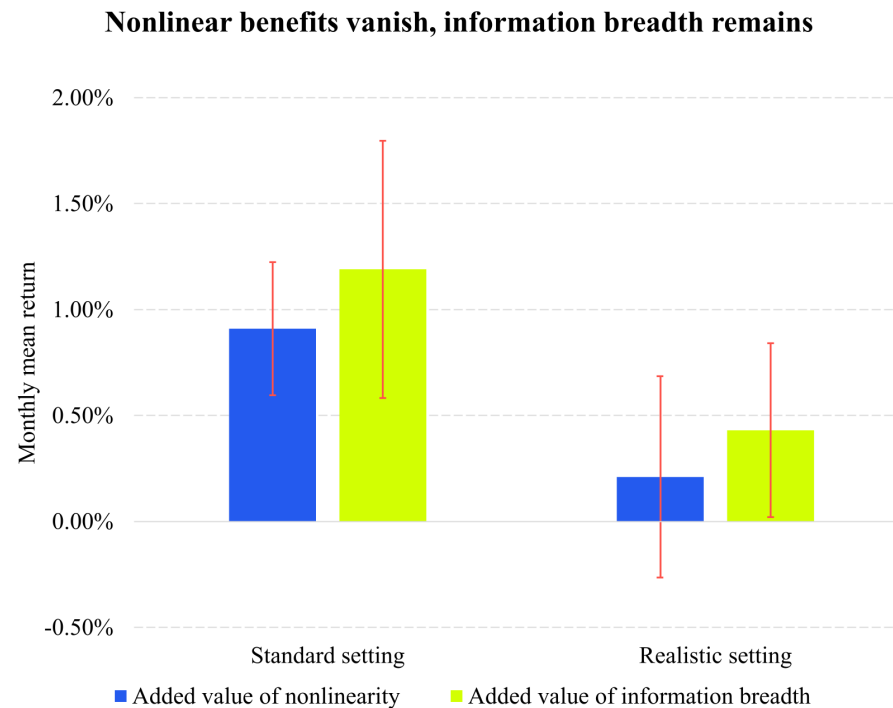
Where does disagreement come from?

- **Political** and **income** differences matter more than **age**, **gender**, **education** and **race** differences.
- Hard news gives less scope for disagreement
- News about social issues ~ disagreement; legal problem news ~ agreement

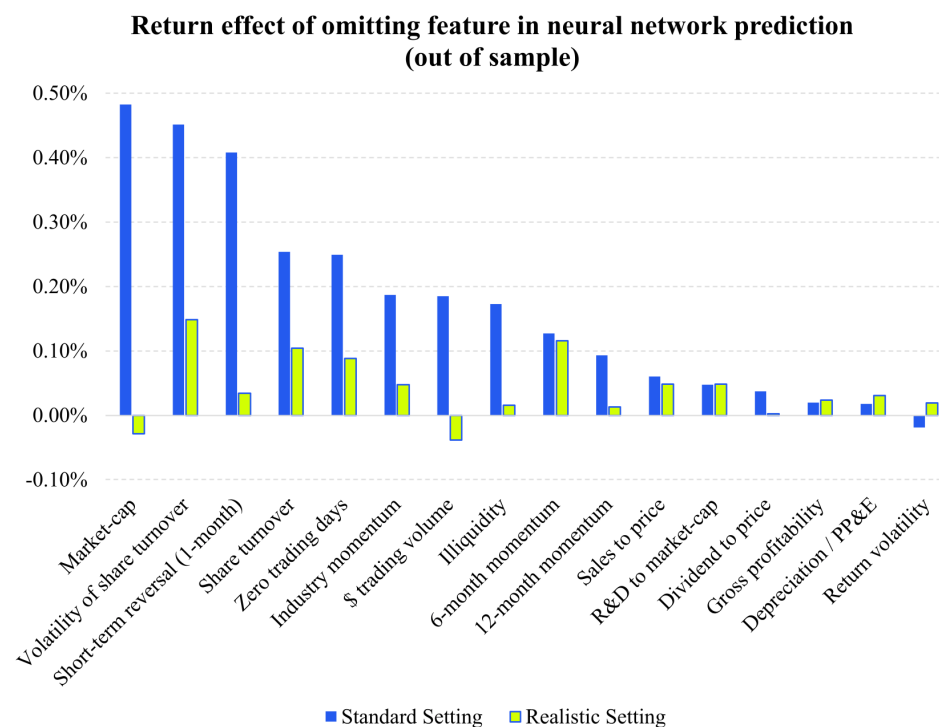
Transaction Costs

Important to Account for Transactions Costs with “Complexity” Grows!

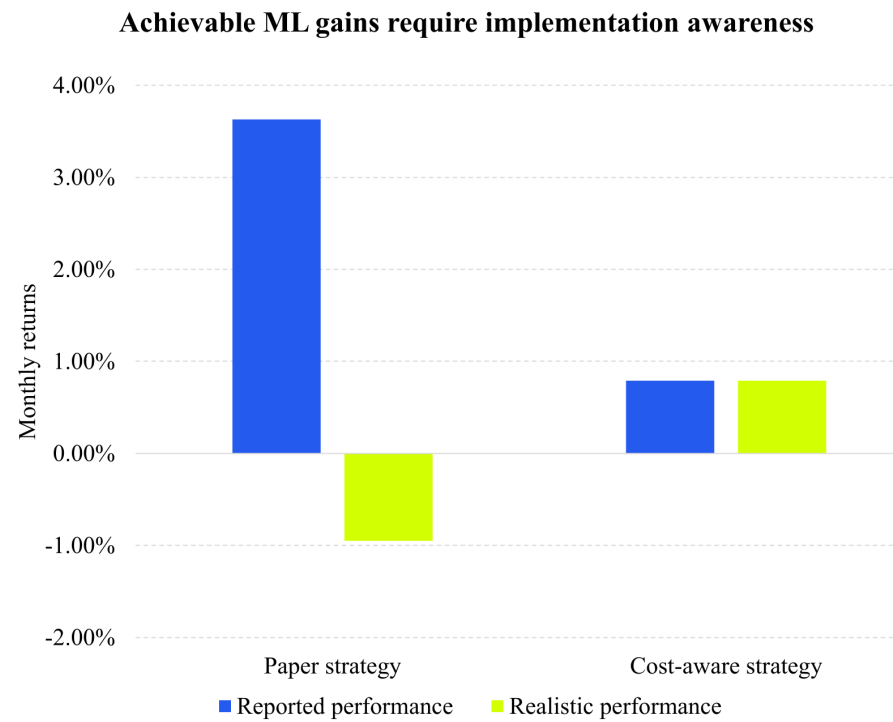
Esakia Goltz 2025



Some Market Factors Imply Implausible Trading



Cost Aware ML Gains



Tax Loss Harvesting

Vanguard

Vanguard research

July 2024

Tax-loss harvesting: Why a personalized approach is important

How TLH works

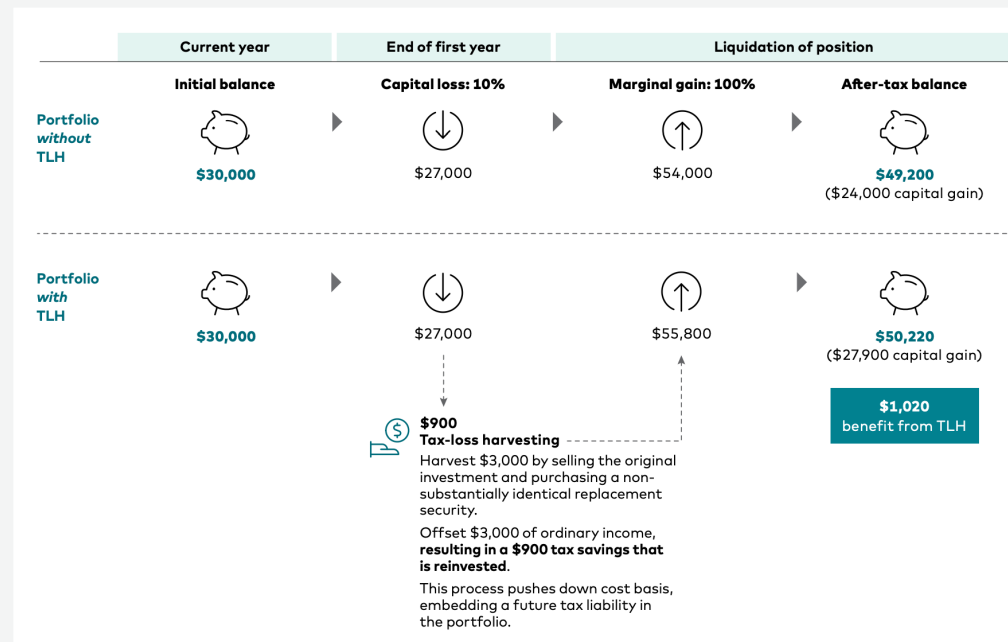
Tax-loss harvesting operates on the principle of converting investment losses into tax savings. Securities held in a taxable account can be sold—or “harvested”—when their value falls below cost basis (the value when they were purchased); the investor thus realizes a capital loss. These capital losses are used to offset an equal amount of capital gains in other parts of an investor’s portfolio and/or up to \$3,000 of ordinary income. If the losses realized exceed the gains realized and \$3,000 of ordinary income in a given calendar year, the extra losses are carried forward and are available to offset capital gains and income in future years.

The following example (**Figure 1**) illustrates how TLH can add value. When the portfolio experiences a loss, the position can be sold, and the captured loss (\$3,000) can be used to offset income, creating a tax savings of \$900. When this \$900 is reinvested in the portfolio, the portfolio grows more than it would have without the reinvestment of tax savings. Ultimately, the TLH portfolio can also generate a larger tax bill because of its lower cost basis; however, this will represent a net benefit if the market appreciates after the reinvestment of tax savings and/or the tax rate paid by the investor is lower at liquidation than at the time of harvest.

Tax Loss Harvesting in Vanguard

FIGURE 1

TLH adds value through tax-rate substitution and increased market exposure



Empirical Analysis of Tax Loss Harvesting Alpha

Financial Analysts Journal | A Publication of CFA Institute
<https://doi.org/10.1080/0015198X.2020.1760064>

Research

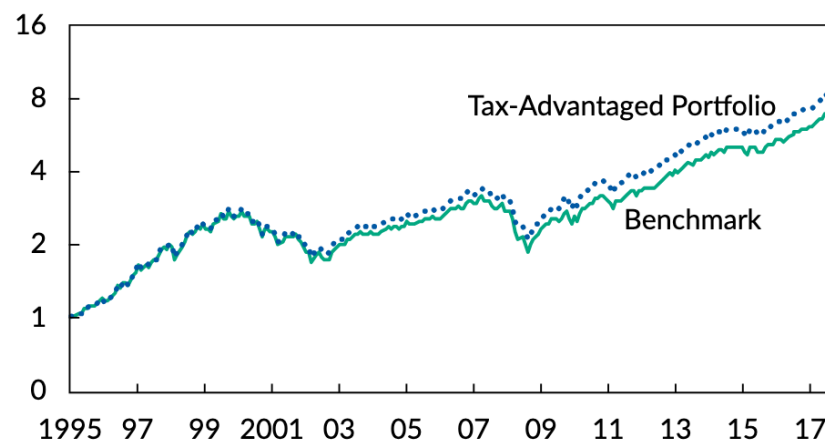
An Empirical Evaluation of Tax-Loss-Harvesting Alpha

Shomesh E. Chaudhuri , Terence C. Burnham , and Andrew W. Lo 

Shomesh E. Chaudhuri is chief technology officer of QLS Advisors, Cambridge, Massachusetts. Terence C. Burnham is an associate professor at George L. Argyros School of Business and Economics, Chapman University, Orange, California. Andrew W. Lo is the Charles E. and Susan T. Harris Professor at Sloan School of Management and director of the Laboratory for Financial Engineering, Massachusetts Institute of Technology (MIT), Cambridge, Massachusetts; Electrical Engineering and Computer Science Department, MIT; Computer Science and Artificial Intelligence Laboratory, MIT; Santa Fe Institute, Santa Fe, New Mexico.

D. 1995–2018

Cumulative Return (%)



LLM Trading?



Operational Performance

"Claude has fundamentally transformed the way we work at NBIM. With Claude, we estimate that we have achieved ~20% productivity gains - equivalent to 213,000 hours. Our portfolio managers and risk department can now seamlessly query our Snowflake data warehouse and analyze earnings calls with unprecedented efficiency. From automating monitoring of newsflow for 9,000 companies to enabling more efficient voting, Claude has become indispensable." - Nicolai Tangen, CEO at NBIM

AI Scaling Meets Trading

- Competing for Alpha:
 - Unlike other AI applications where the value can be broadly shared, trading alpha is zero sum. AI improvements get competed away in arms race
 - Scarce complements are data, infrastructure, latency etc.
 - Leads to “pod shop” economics: stack multiple independent signals because any one signal tends to decay

AI Augmentation

- Operational improvements, improvements in portfolio selection and construction, and research co-pilots can help to augment human performance
- Key caveat is that “performance” in trading often entails accuracy in real world deployment, surviving regime shifts, rather than historical back-fitting

Expanding the Pie

- Personal wealth management is a high-elasticity carveout in this world
 - Tax loss harvesting makes previously expensive services (personalized, daily tax aware rebalancing) nearly free, and demand for this product can scale
- Broader takeaway is:
 - Alpha is harder to sustain and more expensive to find -> markets more efficient, better for passive investing, harder for discretionary stock pickers
 - Highly concentrated quant approaches can reward scale and compute, but value accrues most to scarce complement owners
 - Benefits from broadening access to previously limited asset management